

SA-DAO, Thailand

WMO: 485740

Lat: 6.7892N

Lon: 100.4100E

Elev: 25

StdP: 101.02

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.1	22.8	19.7	14.4	29.2	20.6	15.3	28.4	5.5	30.0	4.9	30.4	0.0	N/A	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.6	35.8	26.3	34.9	26.3	34.2	26.3	27.7	32.8	27.5	32.5	27.2	32.1	2.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.2	30.4	26.2	21.7	30.0	25.9	21.3	29.8	88.9	33.1	87.4	32.6	86.2	32.3	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.0	4.2	3.5	DB	19.7	37.0	1.1	1.6	18.9	38.1	18.3	39.0	17.6	39.9	16.8	41.1	
			WB	19.4	28.7	1.1	0.9	18.6	29.3	17.9	29.8	17.3	30.3	16.5	30.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	27.1	27.7	28.6	29.1	29.0	28.8	28.4	28.4	28.2	28.0	27.5	27.0
	DBStd	1.18	0.98	0.91	1.04	1.08	0.94	0.95	0.91	0.93	0.96	0.79	1.14	1.04
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6629	531	496	577	573	591	563	571	570	546	558	525	528
	CDD18.3	3587	273	262	319	323	332	313	312	312	296	300	275	270
	CDH23.3	37327	2554	2722	3487	3618	3688	3492	3426	3361	3023	2949	2566	2442
	CDH26.7	15045	862	1129	1628	1663	1603	1504	1429	1406	1183	1077	836	726
(14) Wind	WSAvg	0.6	0.8	0.9	0.8	0.5	0.5	0.6	0.6	0.7	0.6	0.5	0.4	0.6
(15) Precipitation	PrecAvg	1794	76	39	100	127	130	124	117	140	158	234	289	245
	PrecMax	2603	366	160	347	282	305	243	211	313	251	433	696	658
	PrecMin	1269	8	1	4	11	57	15	38	34	83	116	116	39
	PrecStd	284	84	43	80	66	49	55	45	58	47	74	139	139
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	34.6	36.9	37.5	36.3	35.1	34.7	34.9	34.5	33.8	33.4	33.0
		MCWB	26.0	24.9	25.9	27.0	27.7	26.7	26.4	26.1	26.7	26.5	26.7	26.5
	2%	DB	32.2	33.6	35.8	35.7	34.8	34.3	33.9	34.0	33.7	33.1	32.5	32.1
		MCWB	25.5	25.2	25.6	26.8	27.1	26.7	26.3	26.1	26.4	26.4	26.6	26.1
	5%	DB	31.6	32.9	34.8	34.5	34.0	33.7	33.2	33.3	33.0	32.3	31.8	31.2
		MCWB	25.3	25.1	25.6	26.6	26.9	26.7	26.3	26.2	26.2	26.4	26.5	25.8
	10%	DB	30.7	32.0	33.5	33.5	33.1	32.8	32.4	32.5	32.0	31.5	30.9	30.3
		MCWB	25.1	25.0	25.5	26.4	26.8	26.6	26.2	26.2	26.1	26.3	26.3	25.6
	0.4%	WB	27.0	26.7	27.3	28.1	28.6	28.0	27.7	27.6	27.5	27.5	27.6	27.2
		MCDB	31.1	32.2	33.6	34.3	34.2	32.5	31.7	32.0	32.2	31.5	31.7	31.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.4	26.2	26.7	27.6	27.8	27.6	27.2	27.1	27.1	27.1	27.1	26.7
		MCDB	30.5	31.6	32.8	33.5	33.0	32.2	31.8	31.9	31.9	31.4	31.1	30.4
	5%	WB	26.0	25.8	26.3	27.2	27.4	27.2	26.8	26.7	26.7	26.7	26.7	26.3
		MCDB	30.1	31.0	32.1	32.9	32.5	32.0	31.6	31.4	31.4	30.9	30.5	29.9
	10%	WB	25.6	25.5	26.0	26.7	27.1	26.9	26.5	26.4	26.4	26.4	26.4	25.9
		MCDB	29.4	30.3	31.6	32.0	32.0	31.7	31.1	31.0	30.9	30.5	30.0	29.4
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	9.7	10.5	9.6	8.8	8.5	8.3	8.4	8.0	7.8	6.7	6.4
		MCDBR	8.9	10.5	12.6	11.0	9.9	9.6	9.5	9.5	9.3	8.9	8.3	8.1
		MCWBR	2.9	3.1	3.4	3.2	3.2	3.0	3.0	3.0	2.9	3.0	3.0	2.8
	5% WB	MCDBR	7.9	9.4	10.6	10.1	9.2	8.9	8.5	8.7	8.5	8.2	7.5	7.0
		MCWBR	3.0	3.2	3.4	3.4	3.4	3.3	3.2	3.0	3.0	3.1	3.1	2.9
(40) Clear-Sky Solar Irradiance	taub	0.457	0.465	0.475	0.469	0.454	0.463	0.467	0.467	0.467	0.457	0.453	0.452	0.459
	taud	2.297	2.248	2.245	2.272	2.337	2.337	2.314	2.301	2.301	2.325	2.338	2.339	2.307
	Ebn at Noon	861	863	855	847	841	822	822	837	857	863	859	851	
	Edh at Noon	134	144	145	139	127	125	129	134	132	130	127	130	
(44) All-Sky Solar Radiation	RadAvg	4.90	5.82	5.87	5.76	5.16	4.95	4.86	4.97	4.84	4.61	4.25	4.14	
	RadStd	0.35	0.35	0.48	0.27	0.27	0.22	0.21	0.18	0.18	0.31	0.33	0.41	

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	+0.22	+0.14	N/A	+0.32	+0.22	+0.19	N/A	N/A	+94

Nomenclature: See separate page